OFFICIAL STARTUP BUSINESS PLAN & COMMERCIAL SPECIFICATION

VETRIX

AI-Powered Veterinary Pharmaceutical Laboratory Information Management System

VETLIMS AI

A governed digital operating system for sample control, quality workflows, compliance evidence, and AI-assisted laboratory decisions.



Faculty: Faculty of Computers and Artificial Intelligence, Menoufia National University

Project type: Graduation Project & Commercial MVP Startup Proposal

Year: 2026

EXECUTIVE SUMMARY

Executive Summary

Vetrix is a veterinary-pharmaceutical laboratory information management system designed to unify sample control, quality workflows, compliance evidence, and AI-assisted decision support in one affordable B2B SaaS platform.

Vision

Make reliable, traceable laboratory operations accessible to veterinary pharmaceutical manufacturers and quality laboratories across Egypt and MENA.

Mission

Replace fragmented paper and legacy workflows with a governed digital system that improves visibility, speed, audit readiness, and data quality.

Problem

- Disconnected records create delays, rework, and weak traceability.
- Smaller and mid-sized laboratories often cannot justify the cost or complexity of legacy LIMS.
- Regulated teams need explainable AI assistance, not opaque automation.

Unified solution

Vetrix combines core LIMS controls, regulatory workflows, and advisory AI with human confirmation at every consequential step.

Visibility

Fewer blind handoffs.

Readiness

Defensible evidence.

Control

People own decisions.

CONFIDENTIAL · Working business-plan document · Product intent and commercial assumptions require validation before external commitments.

MARKET SIZING & DATA

Market Analysis & Target Audience

Vetrix targets regulated veterinary-pharmaceutical operations where sample traceability, quality evidence, and faster release decisions create measurable value.

Manufacturers

Veterinary pharmaceutical manufacturers and in-house QC laboratories.

CROs & testing labs

Contract research organizations and independent testing laboratories.

Regional networks

Distributors and regulated laboratory networks in Egypt and MENA.

TAM

All addressable workflows

Veterinary-pharma and laboratory information workflows in Egypt and MENA.

SAM

Digitally reachable

Manufacturers, QC labs, and CROs requiring cloud or hybrid LIMS capabilities.

SOM

Pilot beachhead

Facilities reachable through direct sales and regulatory partnerships.

All market-size values should be replaced with validated research before external fundraising or pricing commitments.

QUALITY · COMPLIANCE

Regulatory Drivers Create Urgency

Quality teams must demonstrate consistent records, controlled procedures, traceable changes, and accountable approvals. Vetrix is designed to support operational readiness around those expectations.

EDA alignment

Local regulatory workflows and evidence requirements shape the product's regional focus.

GMP alignment

Controlled processes, deviation handling, and auditable records are built into the operating model.

WHO-informed quality culture

Consistent documentation and risk-based oversight guide implementation conversations.

21 CFR Part 11 readiness

Electronic signatures and audit trails are capabilities; final validation depends on customer procedures and deployment controls.

POSITIONING PRINCIPLE

Compliance-enabling, not compliance-substituting.

Vetrix supports a customer's validated quality system; it does not replace that system.

PRODUCT & CAPABILITIES

Product Capabilities & Value Proposition

Vetrix brings the laboratory's operational record, quality controls, and decision evidence into one workflow-native system.

Sample control

Track samples from intake through disposition.

Specification validation

Apply rule-based checks and controlled result review.

OOS workflows

Manage investigations, steps, and approvals.

Governance controls

Use e-signatures, role permissions, and audit trails.

Operational dashboards

See workload, turnaround time, deviations, and release readiness.

VALUE DELIVERED

One operational record reduces handoffs, makes exceptions visible earlier, and gives quality leaders a defensible trail from sample to decision.

AI GOVERNANCE

AI Advisory Suite with Human-in-the-Loop Governance

Every AI capability is advisory. Human users review, confirm, correct, and own the final decision; the system records that intervention for accountability.

PaddleOCR

Document and screen reading for assisted data capture.

Whisper

Hands-free voice entry for controlled workflow inputs.

Pharmacopoeia RAG assistant

Source-grounded answers for guided reference lookup.

Scikit-learn anomaly detection

Signals unusual patterns for human investigation.

Text-to-SQL analytics

Converts natural-language questions into reviewable operational queries.

Supplier risk scoring

Highlights potential supplier concerns using configured evidence and human review.

CONTROL LOOP

Suggest → review → confirm → record

The system accelerates attention without transferring accountability away from qualified people.

MARKET POSITIONING

Competitive Advantage & Market Positioning

Vetrix sits between expensive legacy LIMS and error-prone paper or spreadsheet workflows: more governed than manual systems, and more accessible than enterprise platforms.

USP 01

Regional specialization

Veterinary-pharma workflows are the core focus.

USP 02

Grounded assistance

Pharmacopoeia retrieval supports guided lookup.

USP 03

Pilot-friendly economics

Affordable B2B SaaS with accessible onboarding.

Dimension	Paper / spreadsheets	Legacy LIMS	Vetrix
Traceability	Fragmented	Strong but complex	Strong and workflow-native
AI assistance	None or ad hoc	Limited add-ons	Built-in advisory suite
Regional veterinary focus	Low	Usually generic	Core positioning
Deployment economics	Low upfront / high hidden cost	High implementation cost	Tiered B2B SaaS
Human approval controls	Manual	Available	Explicit in every AI workflow

BUSINESS MODEL

Business & Monetization Model

Vetrix combines recurring subscription revenue with one-time onboarding and integration services. Pricing should be finalized after interviews with at least three target customer profiles and one pilot deployment.

Tier	Best fit	Included direction	Commercial note
Starter	Small QC teams	Core sample tracking, specs, dashboards	Illustrative pricing; validate with pilots
Professional	Growing manufacturers and CROs	Advanced workflows, e-signatures, AI advisory suite	Illustrative pricing; bundle implementation
Enterprise	Multi-site or regulated networks	Hybrid deployment, integrations, governance controls, premium support	Custom quote
Onboarding / integration	All tiers as needed	Data migration, configuration, training, integrations	One-time fee

REVENUE LOGIC

Recurring platform value, supported by implementation depth.

Subscription revenue creates continuity; onboarding and integration services help customers reach a reliable first workflow.

COMMERCIAL LAUNCH

Go-To-Market Strategy

Vetrix should earn adoption through focused workflow proof, credible sector relationships, and pilots that turn operational evidence into an expansion case.

Direct B2B sales

Target quality heads, laboratory managers, plant leaders, and IT owners with a focused workflow demo that shows sample visibility, controlled review, and decision evidence.

Strategic regulatory partnerships

Build credibility through consultants, training providers, and sector networks that understand regulated implementation conversations.

60-day factory pilots

Start with one workflow, baseline turnaround and exception metrics, then expand after evidence is reviewed with the customer.

SALES PROOF POINTS

- Faster sample visibility and fewer manual handoffs.
- Clearer OOS investigation ownership.
- Audit-ready evidence with explainable AI assistance.

OPERATING PLAN

Financial Projections & Operating Metrics

The figures below are illustrative planning assumptions, not historical results or committed forecasts. Replace them with validated pricing, pipeline, staffing, and pilot data.

OpEx categories

- Product and engineering payroll.
- Cloud hosting, security, monitoring, and model usage.
- Implementation, training, customer success, and regulatory support.
- Sales, partnership development, legal, and administration.

MEASUREMENT DISCIPLINE

Validate the model through pilots.

Accounts, revenue, margin, acquisition cost, lifetime value, and churn should become operating measures as evidence accumulates.

Metric	Year 1	Year 2	Year 3
Paying accounts	5	18	45
Revenue	Illustrative: validate	Illustrative: validate	Illustrative: validate
Gross margin	Target range to validate	Target range to validate	Target range to validate
CAC	Track by channel	Track by channel	Track by channel
LTV	Track by cohort	Track by cohort	Track by cohort
Logo churn	Target below 10% annually	Target below 10% annually	Target below 10% annually

Next validation priorities

Confirm target-customer needs, complete pilot discovery, validate tier boundaries, and translate operating evidence into a measured commercial forecast.